Day-Ahead Flight Delay Risk Prediction

Phase 1: Machine Learning

Member 1 Member 2 Member 3 Member 4 Member 5

Advanced Machine Learning Course Project

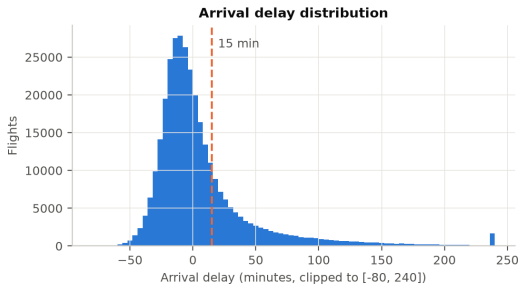
Problem

- Predict on the **night before departure** whether a flight from NYC will arrive **15+ minutes late** (BTS definition).
- Output: a calibrated delay probability, plus an offline what-if rescheduling rule.
- Allowed information at decision time:
  - the published schedule of the day (carrier, route, times, aircraft)
  - outcomes of *strictly earlier* days
- Not allowed: departure delay, actual times, air time, anything from the same day.

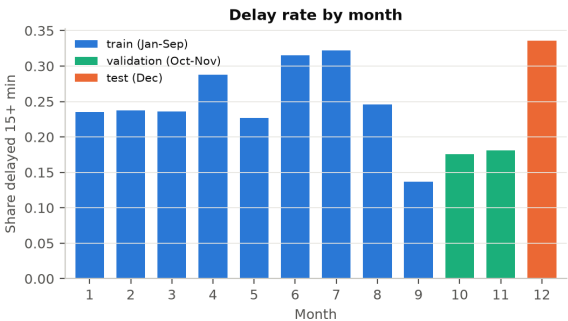
$$P\big(y_i = 1 \mid \{s_j : d_j = d_i\} \cup \{(s_j, y_j) : d_j < d_i\}\big)$$

Dataset: NYC flights 2013

- 336,776 departures from EWR, JFK, LGA
- 16 carriers, 105 destinations, 4,043 aircraft
- 8,255 cancelled + 1,175 diverted $\Rightarrow$ no label, excluded as examples (still counted in scheduled volume)
- 24.5% delayed 15+ min
- corr(dep_delay, arr_delay) = 0.915 $\Rightarrow$ post-departure fields removed

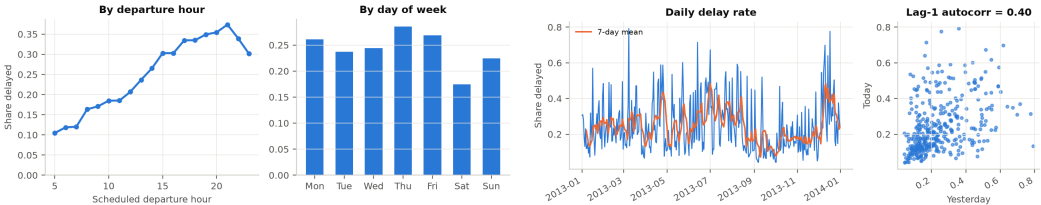


EDA: strong seasonality and a shifted test month



- Chronological split: train Jan–Sep, val Oct–Nov, test Dec
- Base rate: train 0.250, val 0.178, test 0.335
- Test month is the worst of the year, validation among the calmest
- No month / day-of-year features: Oct–Dec never appear in training

EDA: time of day and day-to-day persistence



- Delay rate climbs from 10.5% (05:00) to 37% (21:00): delays propagate through the day
- Lag-1 autocorrelation of the daily delay rate 0.40, lag-2 only 0.13

Feature engineering (35 features)

Calendar	cyclic time of day (dep/arr), cyclic day of week, weekend, days to holiday
Schedule	departure hour, distance, overnight arrival; carrier, origin, dest
Congestion	scheduled flights per origin-day / origin-hour / dest-day / carrier-day; aircraft's legs today and leg number
Long-run history	smoothed past delay rate per carrier, origin, dest, route, carrier-route, origin-hour, flight no., aircraft
Recent history	7-day rates; yesterday's system/origin delay rate, cancellations, mean delay

- Past-only sums $K_g(d) = \sum_{d'<d} k_g(d')$, smoothed: $\hat{r}_g = \dfrac{K_g + m\,p_0}{N_g + m}$, $m = 20$
- **Leakage test:** scramble all outcomes from a cutoff date on $\Rightarrow$ features up to the cutoff must not change

Models and theory

Logistic regression

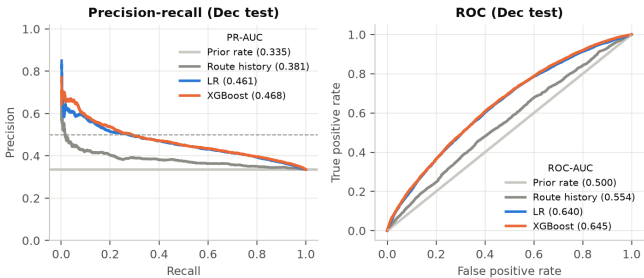
- L2-regularised log-loss, convex
- Intercept optimum $\Rightarrow \sum p_i = \sum y_i$ (calibrated in the large)
- One-hot categoricals, standardised numerics; $C = 10$

XGBoost

- 2nd-order boosting, $g = p - y$, $h = p(1 - p)$
- Leaf $w^* = -G/(H + \lambda)$
- `min_child_weight` $= 2000$: $H \geq 2000$, $h \leq \frac{1}{4} \Rightarrow \geq 8{,}000$ flights per leaf
- depth 4, $\lambda = 20$, 308 trees (early stopping)

Calibration: Platt $q = \sigma(a \operatorname{logit} p + c)$ fitted on Oct–Nov, and a *rolling* Platt refitted daily on the last 28 days (window chosen on November).

Results on December (test)



Model	PR-AUC	ROC-AUC	Brier	ECE
Prior rate	0.335	0.500	0.230	0.086
Route history	0.381	0.554	0.230	0.095
LR	0.461	0.640	0.222	0.103
XGBoost	0.468	0.645	0.221	0.102

- Both models beat route history
- XGB $\approx$ LR on test;
  LR better on validation (0.297 vs 0.276)

Why XGBoost needed heavy regularisation

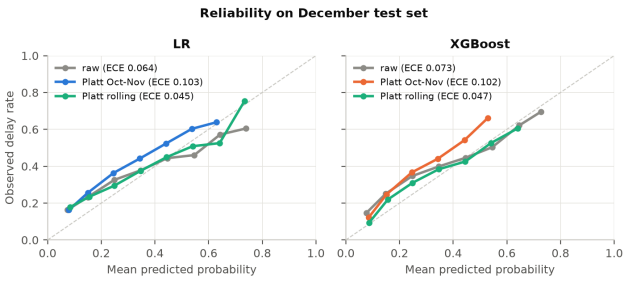
- Train vs validation ROC-AUC: XGBoost 0.752 $\rightarrow$ 0.650, LR 0.710 $\rightarrow$ 0.664
- Day-level features (yesterday's rates) are constant within a day $\Rightarrow$ together they fingerprint the date
- Small leaves let trees memorise the weather of individual training days
- Validation picked the largest `min_child_weight`: every leaf spans $\sim$9 days of flights
- Day-of-year features removed for the same reason

Class imbalance: we tried class weighting

Model	Val PR-AUC	Test PR-AUC	Test ROC-AUC	Test ECE	Mean $\hat{p}$
LR (unweighted)	0.297	0.461	0.640	0.064	0.295
LR, balanced class weights	0.296	0.460	0.641	0.182	0.517
XGBoost (unweighted)	0.276	0.468	0.645	0.073	0.269
XGBoost, positive weight 3.00	0.273	0.466	0.647	0.153	0.488

- 25% of training flights delayed: moderate imbalance
- Weighting barely changes PR-AUC / ROC-AUC
- But it inflates probabilities (mean $\hat{p} \approx 0.5$ vs observed 0.335) and more than doubles ECE
- Kept unweighted; imbalance handled by PR-AUC + precision-based threshold
- No SMOTE: synthetic flights mixing different days break day-level history features

Calibration breaks under the base-rate shift



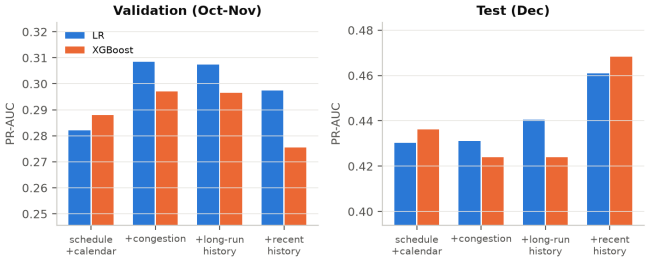
- Oct–Nov Platt learns a low base rate $\Rightarrow$ under-predicts December
- ECE (XGB): raw 0.073, fixed 0.102, **rolling 0.047**
- Label shift theory: only the intercept needs to follow $\log \frac{\pi}{1-\pi}$
- Cost: pooled PR-AUC 0.450

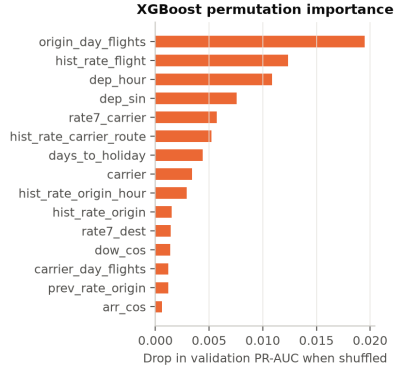
Operating point: precision ≥ 0.5

Model	Threshold	Flagged	Precision	Recall
LR	0.519	1.7%	0.612	0.031
XGBoost	0.498	1.8%	0.661	0.035
LR (rolling cal.)	0.648	0.9%	0.557	0.014
XGBoost (rolling cal.)	0.576	3.6%	0.585	0.062

- Threshold chosen on validation, applied unchanged to December
- Precision holds, recall is low: the shift makes December scores look "too low"
- Reference only (read off the test curve): recall at precision $\geq 0.5 = 0.292$ (XGB), 0.276 (LR)

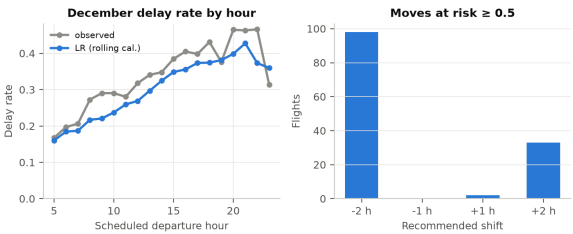
Feature ablation and importance





- Congestion helps most on calm validation months
- Recent history hurts slightly on validation but is the biggest gain on December

What-if rescheduling (December, LR + rolling calibration)



- Risk $\geq 0.5 \Rightarrow$ try ± 1, ± 2 h same day; move if risk drops ≥ 0.05, max 3 moved flights per origin-hour
- 1,979 flagged, 133 moved (98 earlier)
- Moved flights: risk $0.571 \rightarrow 0.510$
- Expected delays among flagged: $1123 \rightarrow 1115$
- Associational, ignores crews, rotations, connections

Limitations and Phase 2

- No weather forecast: day-level weather dominates and caps day-ahead accuracy
- One year, NYC departures only: inbound legs and Oct–Dec seasonality unseen
- Cancelled/diverted flights dropped (cluster on the worst days)
- Rescheduling estimate is not causal

Phase 2: compact MLP with learned embeddings for carrier, origin and destination, same split and features, same metrics.

Summary

- Leakage-free day-ahead pipeline, verified by an automated test
- December PR-AUC 0.461 (LR) / 0.468 (XGB) vs base rate 0.335
- XGBoost only competitive once forced to use large leaves
- Calibration must adapt to drift: rolling Platt cuts ECE from 0.102 to 0.047
- Rescheduling: earlier is better, but gains within ± 2 h are small

Thank you